Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH. CE and IIIrd
Name of the Paper: Building Materials and Construction Equipment
Paper Code: TCE-304
Time: 1.5 hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks)
a. How does Reinforced Cement Concrete (RCC) improve structural strength compared to plain concrete? (CO1)

OR

b. Explain the principle behind Pre-stressed Concrete and its applications in construction. (CO1)

Q2. (10 Marks)
a. Discuss the factors contribute to the sustainability of construction materials? (CO2)

OR

b. Why is there a need for high-performance materials in pavement construction? Explain it. (CO2)

Q3. (10 Marks)
a. How do eco-friendly cements improve the sustainability of concrete construction? (CO2)

OR

b. How does sulfur-extended asphalt differ from conventional asphalt? (CO2)

Q4. (10 Marks)
a. How does the density of Lightweight Concrete differ from traditional concrete, and in what scenarios is it used? (CO1)

OR

b. How does Geo-polymer Concrete offer environmental benefits over traditional Portland cement-based concrete? (CO1)

Q5. (10 Marks)
a. What are the environmental benefits of using Pervious Concrete in construction projects? (CO1)

OR

b. Discuss the benefits of porous asphalt pavement in stormwater management? (CO2)